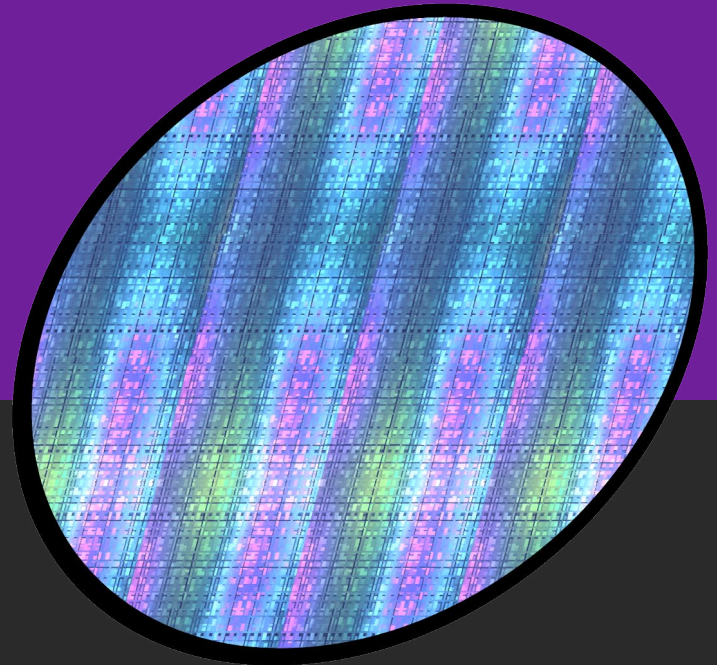


Global Semis

CHIPS IV: China Semis self-sufficiency builds



In 2023, amid the challenged constraints on talent, equipment, and advanced-node chips, we set out our first piece ([report link](#)) answering the questions of what it would take to build a self-sufficient semiconductor supply chain, how much capital might be required and what are the key factors to drive China's success in Semis. In this fourth edition after 3 years, our analysis shows:

- **Ongoing build of supply chain - filling the gaps:** Our **SPE** map shows expansion from Etcher and Deposition to premium models (e.g. high aspect ratio etcher, atomic layer deposition) and more comprehensive categories: Ion Implanter, Inspection, Metrology. **Foundry** towards 7nm and below with chiplet and advanced packaging to drive production yield rate. **AI chips** toward 4 PFLOPS in FP8 (4Q28, [link](#)) from 320 TFLOPS in FP16 (2023), with SuperPods to drive system-level computing power. **Memory** towards LPDDR6 (4Q26, [link](#)) and HBM3 & 3E (4Q26, [report link](#)) from LPDDR4(X) (2024, [report link](#)). **EDA** expands from analog to digital with more tools (e.g. Place and Route; Physical verification) and memory EDA.
- **Ambitious capex targets that amounts to US\$82bn in 2030E, or raised by 79%** to capture the growing demand driven by self-sufficiency, generative AI, and customers' "local for local" strategy. We estimate China 7nm and below logic S/D gap to narrow to 34% by 2035E, and CXMT's supply to capture 50% of China DRAM demand by 2028E, driven by capacity expansion and yield rate ramp-up.
- **What China still needs to do:** China IC volume self-sufficiency was at 70% in June 2026 (vs. 38% in Jan 2010), while the value S/D gap remains wide given lack of Lithography; R&D and Capex gap vs. global leader remains wide despite strong increases, suggesting more capital required to go global. **We introduce China GPU and DRAM TAMs in this report.**
- **Initiating coverage on shares of China's DRAM leader, CXMT, at Buy with a 12M TP of Rmb129:** We expect capacity to more than double by 2030E vs. 2026E to 665k wpm, leading its conventional DRAM supply to reach 41% / 50% of Samsung / SK Hynix by 2028E from 28% / 35% in 2025.

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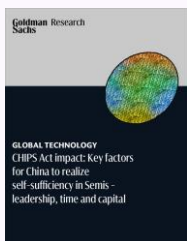
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Explore the series:



CHIPS Act impact

Key factors for China to realize self-sufficiency in Semis - leadership, time and capital



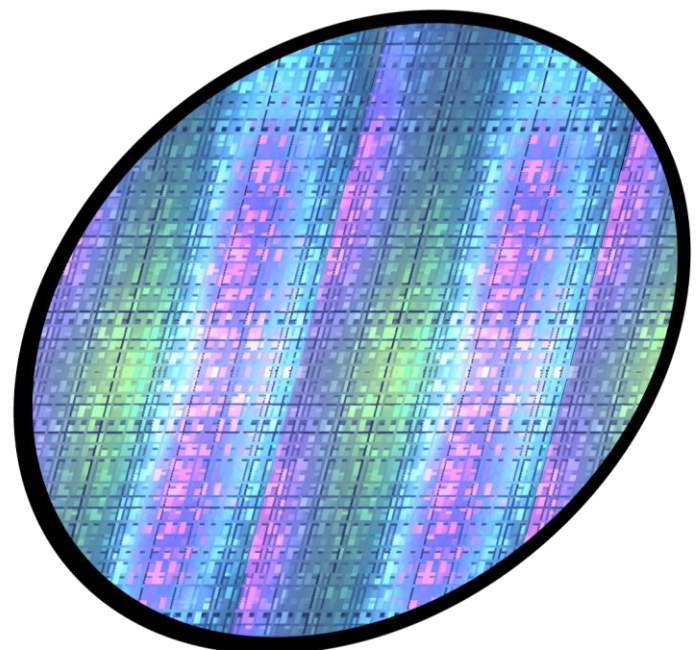
CHIPS Act 2

China's accelerating momentum



CHIPS Act 3

China capex continues to grow: migration ongoing despite challenges

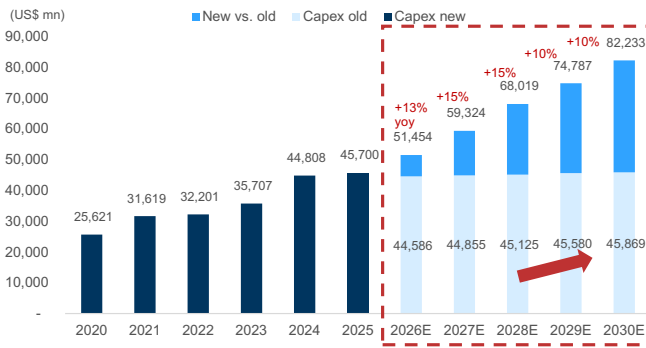


Portfolio Summary: Ongoing build of China Semis

In the first report (2023, [report link](#)), we laid out the pieces of the technology puzzle necessary for China to realize self-sufficiency in semiconductors. In the second piece (2024, [report link](#)), we reviewed China Semis Capex and drivers, where we see mature nodes have expanded more vs. advanced nodes. In the third piece (2025, [report link](#)), we raised China Semis Capex by single digit on both mature nodes and advanced nodes logics and memory, updated local SPE coverage map, and reviewed existing Lithography in China vs. China advanced nodes demand. In this fourth edition (2026), we analyze and update:

- **Product development:** China Semis supply chain continues to develop and fill the gap, driven by (1) generative AI: driving supply chain toward advanced nodes chips design and fabrication, advanced packaging, chiplet, high bandwidth memory (HBM), etc., (2) customers' "local for local" strategy to diversify suppliers to better secure supply as generative AI needs occupy most global-tier suppliers' advanced nodes capacity, and (3) China's ongoing commitments on self-sufficiency. Our SPE map shows expanding from Etcher and Deposition to premium models (e.g. high aspect ratio etcher, atomic layer deposition) and more comprehensive categories: Ion Planter, Inspection, Metrology.
- **Capex:** Strong rise expected at 15-79% to US\$51-82bn in 2026-30E and we forecast China WFE (wafer fabrication equipment) spending to grow 13%/20%/15% YoY in 2026-28E, mainly on higher investments in memory and advanced node technologies to support robust growth in generative AI. We expect this increased capital intensity in China to benefit both global and domestic SPE supply chains.
- **7nm and below supply vs. demand in China:** We project China's 7nm and below wafer demand to grow at a 17% CAGR from 2025 to 2035E, reaching 619k wpm by 2035E. This expansion is primarily driven by AI servers, which we forecast to grow at a 42% CAGR over the same period, on surging demand for generative AI training and inference workloads. We project China's advanced (7nm and below) wafer supply to grow at a 46% CAGR from 2025 to 2035E, reaching 410k wpm by 2035E. Under this scenario, domestic supply and demand gap is projected to narrow to 34% by 2035E (vs. 92% in 2025).
- **Introducing China GPU and DRAM TAMs:** We introduce our China AI chips / DRAM TAM and estimate China AI chips TAM to grow at +142% / +69% / +6% CAGRs in 2025-30E to US\$4,123bn / US\$678bn / US\$66bn by 2030E in our Bull / Base / Bear cases. This implies (1) shipments at 237m / 39m / 4m units, and (2) IT power at 196GW / 32GW / 3GW in 2030E, respectively. Assuming our base case for China AI chips, we estimate China DRAM TAM to grow at +50% CAGR in 2026-28E to US\$257bn, while HBM at +188% CAGR to US\$32bn.
- **Initiating on CXMT at Buy:** Our TP of Rmb129 implies 24x 2027E P/E vs. +77% 2027-28E avg. EPS, driven by AI spending and customers' diversification of supply. We expect capacity to more than double by 2030E vs. 2026E to 665k wpm, positioning CXMT to capture strong demand, while product mix upgrades toward HBM should unlock further opportunities in China's AI market.

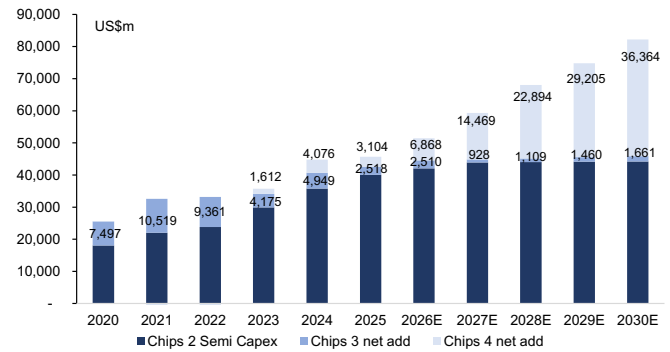
Exhibit 1: Semis capital expenditures in China: we expect it to deliver strong growth and reach US\$82bn in 2030E



Compared to our last estimate in Aug-25, we raised our 2026-30E capex by 15%/ 32%/ 51%/ 64%/ 79%

Source: Company data, Goldman Sachs Global Investment Research

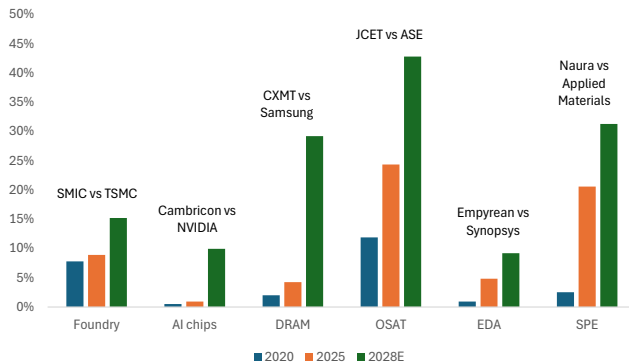
Exhibit 2: China Semi capex changes in Chips 2-4



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 3: R&D: required more capital to catch up global leader / go global

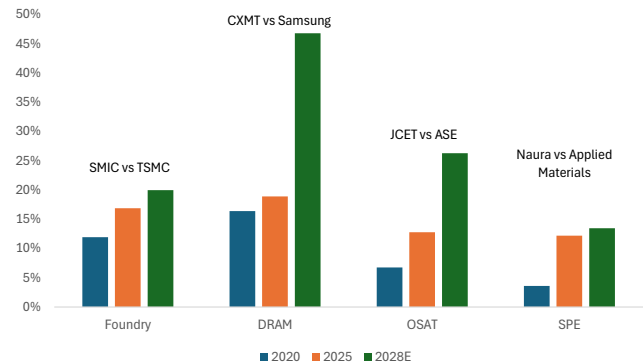
R&D: local leader R&D in 2020/25/28E as a % of global leader R&D in LTM (3Q25-2Q26A)



Source: Company data, Goldman Sachs Global Investment Research

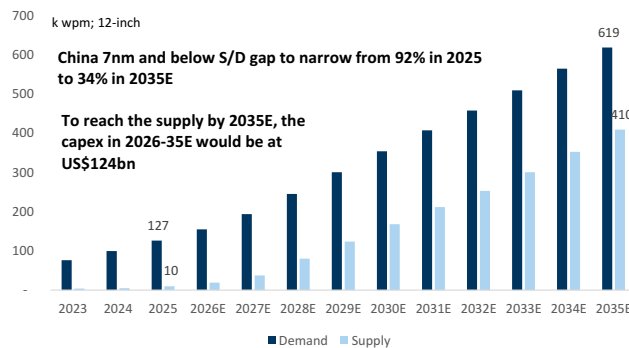
Exhibit 4: Capex: required more capital to catch up global leader / go global

Capex: local leader Capex in 2020/25/28E as a % of global leader R&D in LTM (3Q25-2Q26A)



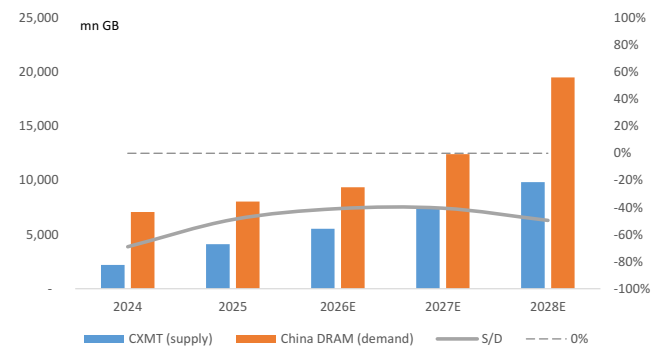
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 5: China 7nm and below supply vs demand



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 6: CXMT supply to cover 50% of China DRAM demand by 2028E



Source: Company data, Goldman Sachs Global Investment Research

China SPE: Increasing product coverage

Expanding product coverage of SPE suppliers in China (Green highlighted are new progress after Aug 2025)

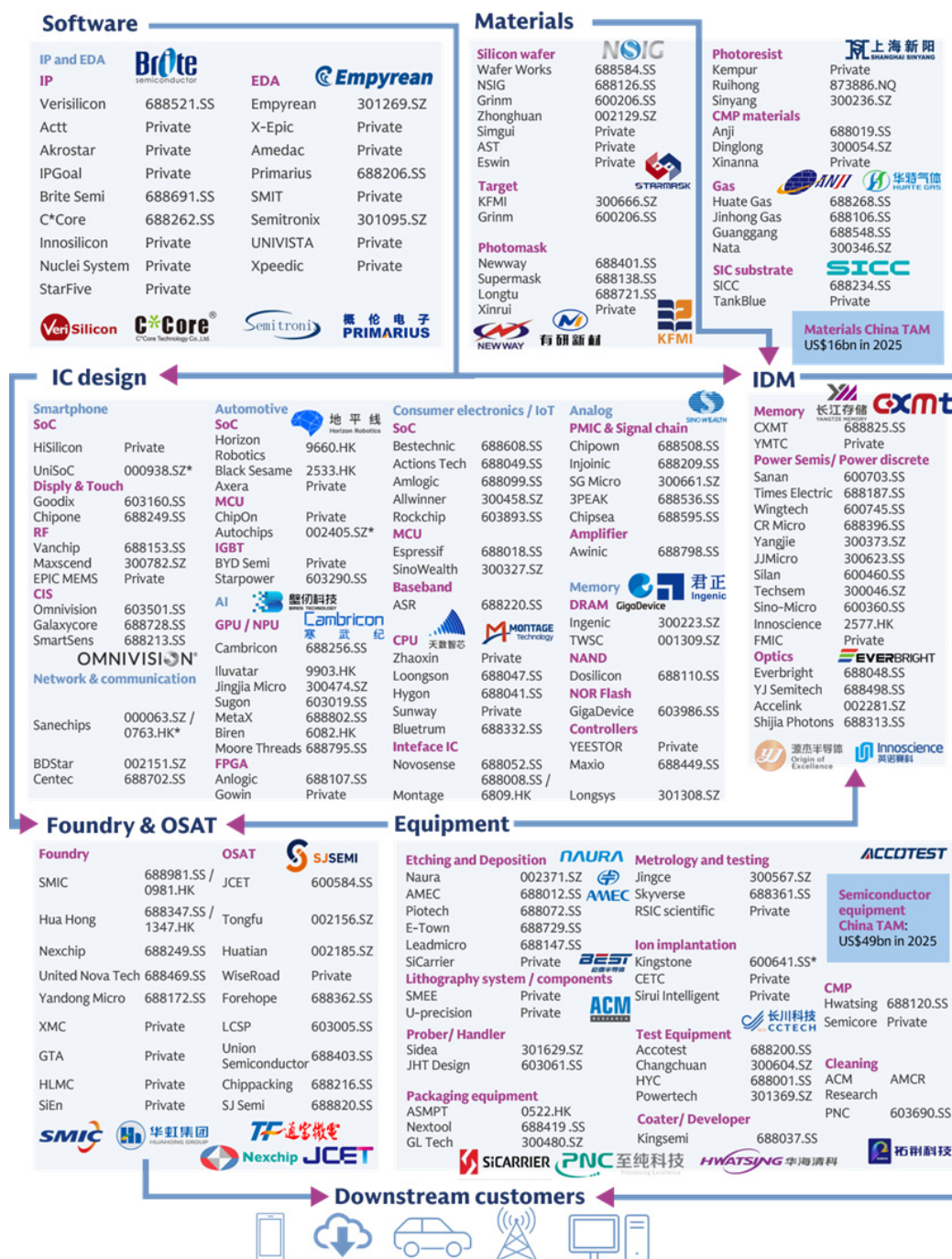
		 NAURA 北方华创	 AMEC	 ACMR ASPIRANT	 KINGSEMI	 PIOTECH	 Vital DeepTech	 SiCarrier	 BEST 北京微电子	 HWATSUNG	 SKYVERSE 中微半导体	 Jingce	 SMEE	 Accotest	 Changchuan 中电	 Leadmicro 微导	 PNC 星维科技	 Power-tech	 Secote	 Autowell	 U-Precision	
SPE segment	Sub-segment	NAURA	AMEC	ACMR	Kingsemi	Piotech	Vital Deeptech	SiCarrier	E-Town	Hwatsung	Skyverse	Jingce	SMEE	Accotest	Chang-chuan	Lead-micro	PNC Process	Power-tech	Secote	Autowell	U-Precision	New product launch
	Ticker	002371.SZ	688012.SS	ACMR	688037.SS	688072.SS	600641.SS	Private	688729.SS	688120.SS	688361.SS	300567.SZ	Private	688200.SS	300604.SZ	688147.SS	603690.SS	301369.SZ	603283.SS	688516.SS	Private	
Front-end																						
Thermal process	Oxidation / Diffusion	✓ (14nm)		✓				✓														
	Annealing	✓ (14nm)		✓				✓	✓ (Advanced)												✓	
	Rapid thermal processing	✓ (14nm)					✓	✓	✓													
	Furnace			✓																		
Litho-graphy	DUV												✓									
	Immersion DUV																					
	EUV																					
Etcher	CCP	✓	✓ (5nm)				✓	✓ (Advanced)														
	Double Damascus CCP	✓						✓ (Advanced)														
	High Aspect Ratio	✓ (50:1)	✓ (90:1 in evaluation)			✓		✓ (Advanced)													2025 AMEC 90:1 high aspect ratio etcher sent to clients for evaluation 2026 Piotech covered etcher product line through acquisition	
	ICP	✓ (7nm)	✓ (5nm)			Evaluation (2026)	✓	✓	✓													
	TSV	✓ (14nm)	✓			Evaluation (2026)																
	Bevel etch	✓	✓	✓																		
	ALE	✓																				
Photo-resist	Coater/ Developer	✓ (Kingsemi)		Evaluation (KrF, 2025)	✓ (28nm)																	
	Remover	✓ (Kingsemi)		✓	✓ (28nm)				✓ (5nm)													
CVD	LPCVD	✓ (14nm)	✓ (Advanced)	✓												✓					2025 Piotech's PECVD Bianca passed clients' evaluation	
	PECVD	✓	✓	✓		✓ (14nm)	✓	✓ (5nm)								✓						
	EPI	✓	Evaluation (2024)					✓ (Advanced)														
	SACVD					✓	✓															
	HDPCVD	✓				✓ (14nm)	✓															
	Flowable CVD	✓				✓																
	MOCVD	✓	✓				✓															
	Tube CVD	✓		✓																		
ALD	PEALD	✓	Announced (2025)			✓ (14nm)		✓ (7nm)								✓					2025 AMEC launched new ALD tools	
	Thermal ALD	✓ (14nm)				✓		✓ (7nm)								✓						
PVD	AI-Pad	✓ (14nm)	R&D (2025-29)			Evaluation (2026)	✓	✓													2026 AMEC's CuBS PVD recognized by multiple leading clients and delivered to a domestic advanced logic device R&D line for evaluation	
	Hard mask	✓ (14nm)						✓														
	CuBS	✓ (14nm)	Evaluation (2026)			Evaluation (2026)	✓ (Matured)	✓														
Ion Implanter	High energy						✓ (28nm)			R&D											2026 Vital Deeptech launched Hyperion advanced-node high-current ion implanter	
	High current	✓					✓ (Advanced)			✓ (Advanced)												
	Medium/ low current						✓ (Advanced)			R&D												
CMP	Memory		✓ (Advanced)							✓ (Advanced)											2026 AMEC covered CMP product line through acquisition	
	Logic		✓							✓ (14nm)											2026 AMEC covered CMP product line through acquisition	

SPE segment	Sub-segment	NAURA	AMEC	ACMR	Kingsemi	Piotech	Vital Deeptech	SiCarrier	E-Town	Hwatsing	Skyverse	Jingce	SMEE	Accotest	Chang-chuan	Lead-micro	PNC Process	Power-tech	Secote	Autowell	U-Precision	New product launch
	Ticker	002371.SZ	688012.SS	ACMR	688037.SS	688072.SS	600641.SS	Private	688729.SS	688120.SS	688361.SS	300567.SZ	Private	688200.SS	300604.SZ	688147.SS	603690.SS	301369.SZ	603283.SS	688516.SS	Private	
Front-end																						
Cleaning tool	Single wafer	✓ (14nm)		✓ (28nm)	✓					✓							✓					2026 Kingsemi's single wafer cleaning tool passed clients' evaluation and received repeated orders
	Batch	✓ (28nm)		✓ (40nm)													✓					
Electro-plating	ECP			✓ (28nm)																		
Inspection	Bright field pattern wafer		✓ (Raintree)					✓			Evaluation (2024)	✓ (14nm)										2026 Jingce's dark field pattern wafer inspection tool completed the R&D stage
	Dark field pattern wafer							✓			✓	✓ (28nm)										
	Unpatented wafer							✓			✓	✓ (Advanced)							✓			
	Patterned wafer										✓											
	EBI		R&D (2025)									✓ (14nm)										
	DRSEM		R&D (2025)																			
Metrology	Optical critical dimension (OCD)		✓ (Raintree)								✓	✓ (7nm)										2026 Skyverse launched first CESEM tool
	CDSEM		R&D (2025)								Announced (2026)	✓ (7nm)										
	Overlay accuracy							✓			✓ (Matured)											
	Dielectric thin-film		✓ (Raintree)							Announced (2026)		✓ (Advanced)										
	Metal thin-film		✓ (Raintree)								✓ (Matured)	✓ (Advanced)										
	3D Metrology										✓ (Advanced)											
Back-end																						
Wafer grinding/thinning	Wafer grinding/thinning									Announced (2026)												2026 Hwatsing shipped its wafer grinding/thinning tool
Testing	Analog													✓	✓			✓				
	SoC											Announced (2026)		✓	✓			✓				2026 Jingce launched its JH5900-A SoC testing tool
	Power							✓						✓	✓			✓				
	Memory											✓			✓			✓	✓			2026 Powertech covered memory testing through acquisition
Electroplating	ECP	✓		Evaluation (2025)																		
Die bonder	Traditional				✓ (Temporary bonding)															✓		
	TCB																					
	Hybrid bonding	Announced (2026)				✓															✓	2026 NAURA launched its hybrid bonding tool

Part 1: Ongoing build of supply chain - filling the gaps

(1) China Semis: Leaders and various suppliers flourishing on multiple fronts

Exhibit 7: China's semiconductor ecosystem



* ticker of parent company

Source: Company data, Goldman Sachs Global Investment Research

(2) Toward advanced nodes: 7nm and below S/D in China

Demand: We model China's 7nm and below wafer demand across eight end markets, applying three key assumptions to each: end-device shipments, advanced-node (7nm and below) penetration rates, and die size (which dictates chips per wafer). Based on this framework, we project China's 7nm and below wafer demand to grow at a 17% CAGR from 2025 to 2035E, reaching 619k wpm by 2035E. This expansion is primarily driven by AI servers, which we forecast to grow at a 42% CAGR over the same period, on surging demand for generative AI training and inference workloads. **Supply:** We model China's 7nm and advanced wafer supply, applying two key assumptions: capacity expansion plans and yield rate ramp-up trajectories. Based on this framework, we project China's advanced (7nm and below) wafer supply to grow at a 46% CAGR from 2025 to 2035E, reaching 410k wpm by 2035E. This growth is primarily driven by SMIC's aggressive capacity expansion (adding 30-50k wpm annually from 2026-31E, and 20k wpm per year from 2032-35E) alongside steady yield rate ramp up (rising from 23% in 2026E to 50% in 2030E, and reaching 75% by 2035E). Under this scenario, domestic supply and demand gap is projected to narrow to 34% by 2035E (vs. 92% in 2025).

Exhibit 8: China market 7nm supply vs. demand

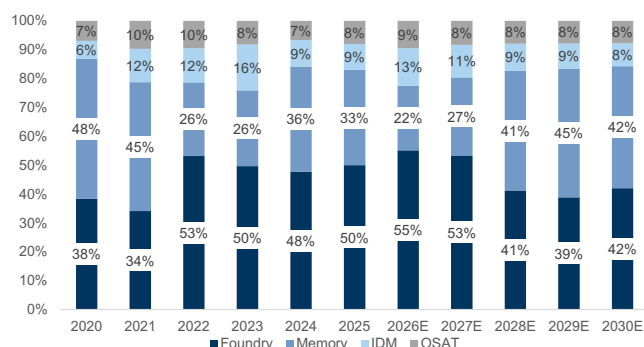
		2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2025-35E CAGR
Demand - China advanced logic (7nm and below)															
Total wafer demand, 12-inch	k wpm	76	99	127	155	194	246	301	354	408	458	510	565	619	17%
Smartphones	k wpm	27	31	33	32	35	38	41	41	42	43	44	44	44	3%
PCs	k wpm	6	6	6	6	6	6	6	6	6	7	7	7	7	1%
Servers	k wpm	5	6	7	10	15	25	35	45	56	70	85	101	120	32%
IoT	k wpm	0.2	0.3	0.3	0.4	0.7	1	1	2	2	3	3	3	3	27%
Smart wearables	k wpm	2	3	4	5	7	9	12	15	18	21	25	28	31	22%
Automotive electronics	k wpm	2	4	5	6	8	10	13	16	18	21	24	26	26	18%
Industrial electronics	k wpm	30	45	64	87	114	144	179	212	245	272	300	330	360	19%
Telecom/ Networking infra	k wpm	4	5	7	8	10	12	14	17	19	22	24	26	28	15%
Supply - China advanced logic (7nm and below)															
Total wafer supply, 12-inch	k wpm	4	5	10	19	37	80	124	168	212	253	301	353	410	46%
SMIC	k wpm	3	4	8	15	29	60	92	123	151	177	205	235	266	43%
Hua Hong	k wpm	-	-	-	0	1	5	9	15	23	32	45	60	77	100%
Others	k wpm	1	1	2	4	7	15	23	31	38	44	51	59	67	43%
Mix	%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	
SMIC	%	80%	80%	80%	79%	78%	75%	74%	73%	71%	70%	68%	66%	65%	
Hua Hong	%	0%	0%	0%	1%	2%	6%	8%	9%	11%	13%	15%	17%	19%	
Others	%	20%	20%	20%	20%	20%	19%	18%	18%	18%	17%	17%	17%	16%	
S / D															
S / D ratio		-95%	-95%	-92%	-88%	-81%	-67%	-59%	-53%	-48%	-45%	-41%	-38%	-34%	

Source: Company data, Goldman Sachs Global Investment Research

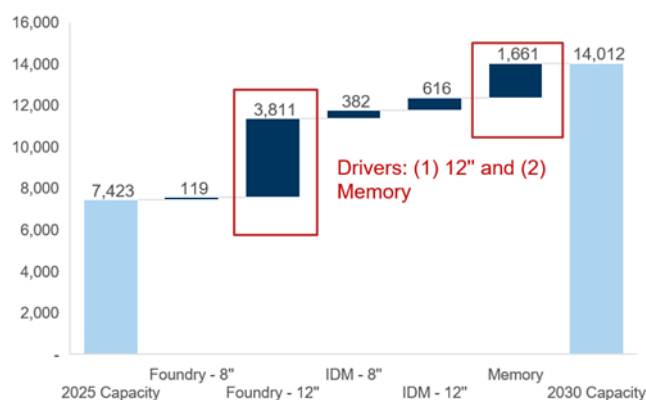
Part 2: Ambitious capex targets that amounts to US\$82bn in 2030E

(1) China Semis Capex: AI to drive spending and technology migration

Semiconductor capital expenditures in China was flat in 2025, post the double digit increase in 2023-24, and we expect double-digit growth to resume through 2030E at +13% / +15% / +15% / +10% / +10% YoY in 2026-30E to US\$82bn by 2030E, driven by memory and advanced nodes capacity expansion under the growing generative AI trend and as global design houses leverage more on local supply chains. The majority of capex will stem from foundries and memory players, which combined, will account for 80-85% of the capex in the next few years in China.

Exhibit 9: Semis capital expenditures in China: foundry and memory players as key drivers ahead

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 10: China will add 6m wpm of 8"/12" capacity in 2025-30E, increasing from 7m wpm to 14m wpm

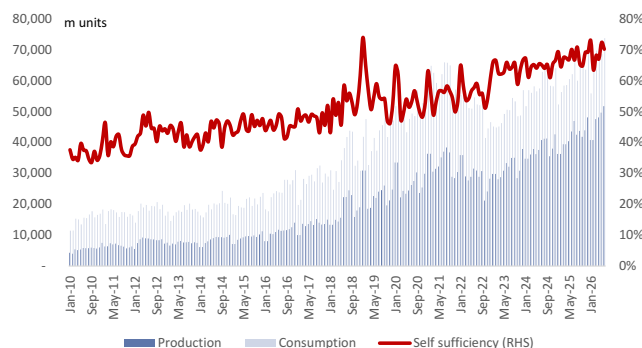
Source: Company data, Goldman Sachs Global Investment Research

While some investors have been more skeptical and assume that China will see a near-term slowdown in capacity additions after the rapid ramp up in 2023-24, we remain positive on growing semis capex ahead due to:

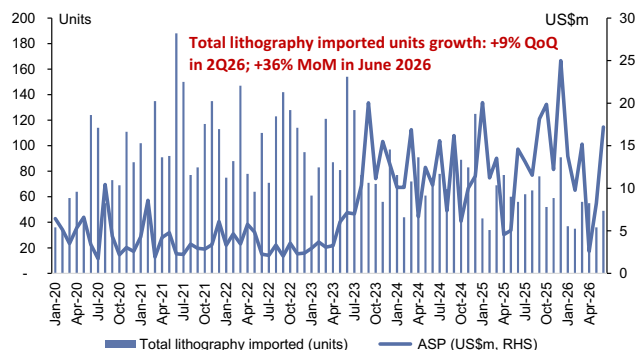
- **Growing foundation models and AI applications will drive generative AI demand** in China, along with the industry's strong commitment to developing the local ecosystem, thereby driving technology migration across AI chips, foundry, memory, advanced packaging and supernodes development. Huawei Supernodes is migrating from connecting 384 AI chips in 2025, to 1,024 AI chips in mid 2026, to 8,192 AI chips in 4Q26, and to 15,488 AI chips in 4Q27, along with specification upgrades of each generation of AI chips. The product development across the supply chain reaffirms our confidence in Semis Capex investment in China, and AI requires premium specifications (i.e. advanced nodes, advanced packaging, high-end memory), consuming more capex per IC. CXMT listed in China and raised Rmb58bn (US\$8bn) in Jul 2026 ([link](#)) and multiple fabless AI chips have also listed in the past 12 months: MetaX (Dec 2025), Moore (Dec 2025), Biren (Jan 2026), Iluvatar (Jan 2026), with additional issuers in the pipeline.
- **Diversification of production sites:** global-tier fabless companies adopt a "local-for-local" strategy to reduce macro impacts (e.g. FX, pandemics, geopolitical tensions), driving Semis Capex investment in China. Moreover, China is a major market globally, consuming ~30% of global semis demand.
- **Generative AI consumes global-tier suppliers' production**, offering more opportunities for Chinese suppliers in conventional ICs. This transition to local ICs / memory adoption is also heavily supported by Chinese brands which are major contributors globally across key sectors such as smartphones (e.g. Huawei, Xiaomi, OPPO, Vivo, Transsion), PC (e.g. Lenovo), TV (e.g. TCL, Hisense, Xiaomi, Skyworth), and automotive (e.g. BYD, Geely).

Exhibit 11: Volume gap narrowing; value gap could be the next to narrow

China IC production volume vs China IC consumption volume



Source: WIND

Exhibit 12: Total lithography imported: shipment and ASP trend


Source: China Custom

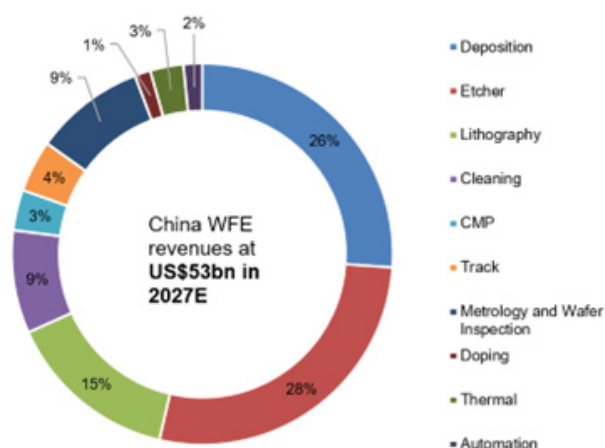
From a capex perspective, we estimate there will be a build out of 7m in new capacity in China over 2025-30E (measured in 8” wafer equivalent). We estimate semiconductor capacity in China (based solely on 8” and 12” lines) was 6m wpm in 2025, and will increase to 14m by 2030E, primarily driven by capacity expansion of advanced nodes and high-end memory chips. These technologies have higher technical requirements and need more sophisticated manufacturing equipment with relatively high ASP, which should continue to drive the semi capex trend in China.

(2) China Semis Capacity: Opportunities for both local and overseas suppliers

We expect WFE (wafer fabrication equipment) revenues in China to grow by 20% to US\$53bn in 2027E, riding on growing semiconductor spending. China’s WFE revenues will account for 29%~31% of global WFE spending in 2026-28E, compared to only 22% in 2022. The growing spend in China for new capacities benefits global and local equipment suppliers. Chinese equipment suppliers account for US\$10.2bn/ 13.5bn/ 17.8bn/ 22.9bn of China’s WFE market in 2025-28E, based on our bottom up forecasts. **The local suppliers produced 26% of the equipment in 2025 (measured by value), which we expect will increase to 38% by 2028E.** On the other hand, global SPE suppliers remain the key contributors to the market with their leading technologies and IPs.

Exhibit 13: China WFE market will be US\$53bn in 2027E

Deposition, Etcher, Lithography are the largest markets



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 15: Chinese players will account for 31% / 34% / 38% of the total WFE revenues in China in 2026-28E

WFE revenues in China, local vs. overseas players



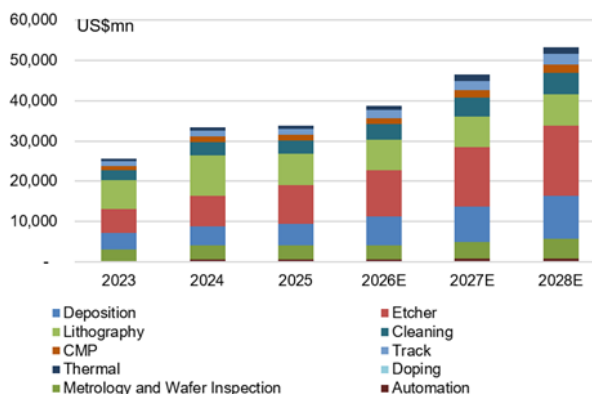
Chinese companies share = Local revenues of Chinese WFE companies / China total WFE revenues

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 14: China WFE spending model

China WFE spending (US\$mn)	2023	2024	2025	2026E	2027E	2028E
GSe China WFE spending (US\$mn)	29,926	38,850	39,080	44,251	52,951	60,685
YoY %	47%	30%	1%	13%	20%	15%
Deposition	7,124	8,781	9,361	11,208	13,830	16,339
NAURA	839	1,496	1,602	2,229	3,036	3,796
AMEC	-	22	71	235	336	431
Etcher	5,927	7,653	9,637	11,533	14,565	17,362
AMEC	658	1,018	1,375	1,784	2,414	3,083
NAURA	764	1,104	1,602	2,229	3,036	3,796
E-town	67	58	105	249	389	462
Lithography	7,283	10,003	7,822	7,638	7,759	7,960
Cleaning	2,380	3,315	3,205	3,773	4,623	5,183
ACMR	404	579	626	787	1,192	1,436
Naura	93	130	193	278	376	499
Hwatsing	-	4	20	35	35	35
CMP	1,010	1,310	1,444	1,562	1,825	2,062
Hwatsing	319	408	517	599	707	860
Track	1,303	1,593	1,440	1,887	2,331	2,637
Metrology and Wafer Inspection	3,214	4,117	4,043	4,179	4,973	5,692
Doping	734	760	501	594	714	770
Thermal	653	747	963	1,148	1,496	1,731
NAURA	112	158	209	261	313	366
E-town	47	86	207	246	400	469
Automation	299	572	666	729	837	950
China WFE spending Total (US\$m)	29,926	38,850	39,080	44,251	52,951	60,685
China semiconductor capex (US\$m)	35,707	44,808	45,700	51,454	59,323	68,018
Equipment to total semis capex	84%	87%	86%	86%	89%	89%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 16: China WFE spending trend by products

Source: Company data, Goldman Sachs Global Investment Research

(3) Implications for Global SPE**Implications for US SPE companies (covered by Jim Schneider)**

We expect US suppliers to continue to maintain a strong, but lower market share position in domestic China WFE spending over time. We see this being driven by two main factors: (1) incremental US export controls, which drove a downtick to sales starting in 2H25; (2) increased focus from domestic CapEx spenders on local WFE

providers, as we outline above. However, we expect China shipments from US providers to remain at the upper end of guidance for mid-20% of revenue over the long run given the steady China CapEx trends we expect – and we note that most major US SPE companies have again revised up their China shipment estimates to reflect modest growth in 2026.

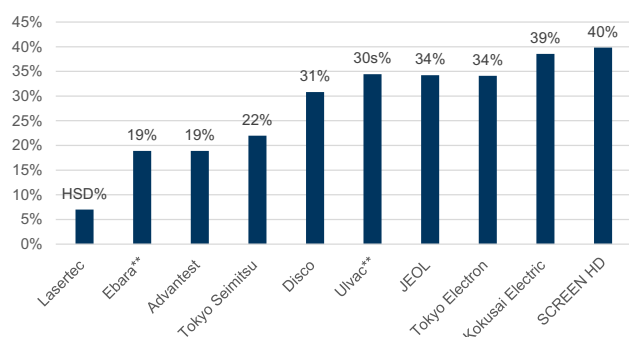
We have a very constructive view of global CapEx growth for multi-national SPE providers as a whole, with global WFE likely up 36% in 2026 and 45% in 2027, based on the latest revisions to our numbers. We expect to see outsized growth for etch and deposition equipment given outsized growth in Gate-All-Around (GAA) structures in Logic, stacked DRAM/NAND structures in Memory, and Advanced Packaging. As such, in our US coverage, we have a relative preference for Applied Materials (AMAT, Buy), Lam Research (LRCX, Buy), and Onto Innovation (ONTO, Buy).

Perspective from the Japan SPE names (covered by Shuhei Nakamura)

The China semiconductor market continues to be an important market for the Japanese SPE companies. Revenues to China for the four major front-end equipment companies we cover (Tokyo Electron, SCREEN, Kokusai, and Ebara) accounted for 35% of their SPE revenues in FY2025 (FY3/26 or FY12/25), or Y1.21 trillion in absolute terms. This figure was down from the previous fiscal year in FY2024 when China accounted for 41% of their SPE revenues, or Y1.44 trillion in absolute terms, likely as a result of 1) pull-in demand ahead of the possibility for further export restriction, and 2) facing more competition from the local equipment companies in China.

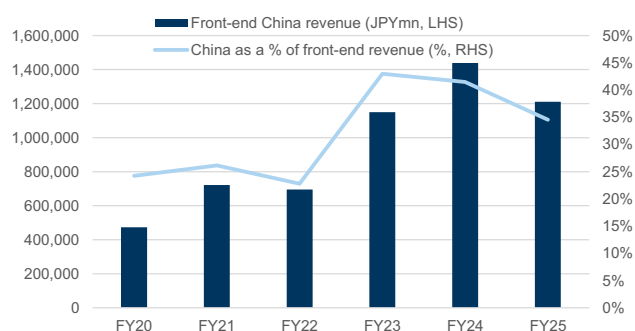
That said, all four companies are expecting their China revenues to be going up in absolute terms for FY2026 and beyond. Aside from the fact that China continues to see strong capex appetite from major semiconductor companies (e.g., SMIC, Hua Hong, CXMT, YMTC), there seems to be less pressure from the local equipment companies in terms of market share dynamics, as the focus of the investment turns to more leading-edge nodes vs. mature nodes. Non-China equipment companies, including the Japanese ones, are getting strong orders from the Chinese customers as China fabs are trying to ensure better yield for the leading edge processes. Hence, we are currently expecting China revenues to rise for all four front-end equipment companies we cover in FY2026-28, although % of China revenues would moderately decline as non-China portion of the revenues would be growing faster than China revenues.

On the back-end side, the three companies we cover (Disco, Advantest, and Tokyo Seimitsu) have all seen a steady increase in China revenues in absolute terms over the years. As China attempts to build their own AI ecosystem, we have seen strong investment appetite from a number of China OSAT players. We continue to believe China will be an important market for the back-end equipment players as well, absent any major export controls going forward.

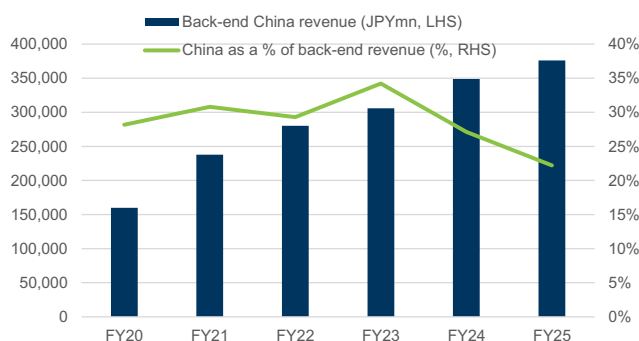
Exhibit 17: China as a % of sales for our SPE coverage names (FY0)

** Uses CW-revenue (which includes non-SPE businesses)

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 18: China revenues (and % of SPE sales) for the four front-end equipment companies we cover

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 19: China revenues (and % of SPE sales) for the three back-end equipment companies we cover

Source: Company data, Goldman Sachs Global Investment Research

Implications for US SPE companies (covered by Alexander Duval)

ASML (ASML.AS, Buy on CL): We reiterate our Buy rating (on CL) on ASML and continue to see the company as a beneficiary of strong China domestic lithography demand as the region works towards developing further self-sufficiency in its semis production. As such, ASML expects the region to contribute c.20% of group revenues in 2026 which is towards the upper end of its historical range of 15-20%. That said, we believe that China revenues will maintain a strong trajectory in future years driven by the construction of new fabs as it digests the capacity of lithography machines shipped in recent years. We also see scope for Chinese memory players to gain share in western markets in the more basic side of the market (given tight supply-demand conditions in conventional memory), further supporting lithography spend by Chinese producers. More broadly, ASML continues to ramp capacity on EUV and is investigating scope for 110 EUV units per year in 2028. We continue to see scope for ASP increases as EUV machines continue to deliver higher throughput and performance.

ASMI (ASMI.AS, Buy): We reiterate our Buy rating on ASM International and note the company continues to see strong demand for its Deposition tools from mature Logic/Foundry applications in China, per 2Q26 results. In this vein, we highlight that the

company reported strong orders from China in 1H26 and the visibility on demand for 2027 appears solid (albeit it is dependent on customer behaviour and the regulatory environment). More broadly, we see China as a key region for ASM's topline and gross margin trajectory given the sales to this region are margin accretive.

BESI (BESI.AS, Buy): We reiterate our Buy rating on BE Semiconductor as it continues to benefit from robust additions of advanced packaging capacity to support 2.5D CoWoS, photonics and pluggable as well as a recovery in high-end smartphone modules. Further, we note that the company pointed towards initial signs of a recovery in industrial and automotive demand in the region which could support the company's traditional packaging portfolio in our view.

Valuation and key risks: ASML: We are Buy-rated on ASML with a 12-month price target of €2,200 based on a 32x 2HCY27+1HCY28E P/E multiple. Key risks to our view and price target include EUV delays, capex cyclicality and unfavourable market share shifts. **ASMI:** We are Buy rated on ASMI with a 12-month price target of €955 based on a 20x 2HCY27+1HCY28E EV/EBITDA multiple. Key risks to our view and price target include a worsening of the semi cycle, stronger-than-expected competition and high customer concentration. **BESI:** We are Buy rated on BESI with a 12-month price target of €325 based on a 29x 2HCY27+1HCY28E EV/EBITDA multiple. Key risks to our view and price target include customer spend cyclicality, hybrid bonding adoption delays and increasing competition.

CXMT – Implications for global memory (covered by Giuni Lee, Shuhei Nakamura, and Jim Schneider)

Regarding CXMT's sales to overseas (including US-based) customers, we believe large US customers are highly motivated to use all available suppliers, including CXMT – particularly for smartphone and PC applications despite CXMT still being a few generations behind in technology node compared to its global memory peers. This desire is motivated by very high conventional DRAM prices as well as limited supply, as most global DRAM makers have shifted production toward HBM products – and current high price levels have resulted in both unit demand destruction and meaningful margin compression for many OEM customers. As a result, we believe several US customers are likely to qualify CXMT product (mobile DRAM and conventional DRAM) for their smartphone and PC products. We do not expect CXMT to make inroads into US-based customers with its emerging HBM product portfolio in the near-to-medium term, given its early stage of technology maturity in this area.

With respect to volume shipments, we believe several exogenous factors will impact CXMT's ability to ship in large volumes to US-based customers. The most important of these factors is geopolitical: as we have written extensively, there has been a long history of import and export restrictions on semiconductor technology and products between China and the US. We believe the governments of both the US and China will carefully weigh potential restrictions on memory in the context of the broader restrictions on other products including GPUs, semiconductor capital equipment, and EDA software, among others. We will be closely monitoring the status of upcoming US/China political summits and other trade discussions in order to gauge the forward direction of any incremental restrictions. More broadly, we expect CXMT's very aggressive capacity investment plan to be one of the important factors which global investors consider when weighing industry supply/demand balance in 2028, where currently our base case is a

continued DRAM supply tightness during the year (although less tight compared to 2027).

Korea memory (covered by Giuni Lee)

The rapid expansion of China's AI market, driven by the push to catch up with U.S. frontier models, is fueling higher spending on domestic semiconductor capex. This massive infrastructure build out is poised to serve as an opportunity for the memory manufacturers.

We owe this opportunity to the structural shift in AI workloads that has elevated memory to a critical component for AI model performance and efficiency. As massive compute capabilities of modern AI accelerators far outpace conventional memory delivery speeds, memory bandwidth and capacity have become the primary bottlenecks for scaling LLMs and agentic AI, driving a sustained, long-term demand for advanced memory solutions. Accordingly, our global team expects the China DRAM market to grow at a 50% CAGR in 2026-2028E to US\$257bn, and reach close to 30% of the global DRAM market TAM. Specifically, China HBM market to grow at a 188% CAGR during the same period to US\$32bn.

Against this backdrop, CXMT has increased its presence by primarily focusing on satisfying domestic demand, while export usage is likely to be driven by both commercial and geopolitical considerations. The company could also see an increase in its global presence via Chinese brands' global expansion.

Even with substantial projected growth, we forecast CXMT to cover only 50% of China's DRAM demand and 40% of its HBM demand by 2028. In addition, there is a technological gap of around 2-3 generations for the company as it mainly produces 1z node compared to the global memory suppliers including Samsung Electronics (SEC) and SK Hynix (Hynix) that are transitioning from 1a/1b node to 1c node. With Chinese semiconductor companies currently having limited access in procuring WFEs including EUV tools due to restrictions, we believe the technology gap is unlikely to narrow soon.

As a result, a large portion of the rapidly expanding China DRAM demand will inevitably continue to rely on the global memory suppliers, and we believe despite the rising competition from CXMT, both SEC and Hynix will continue to have significant opportunities in the China market.

Part 3: China GPU TAM

(1) Demand: US\$678bn / US\$4tn by 2030E in our Base / Bull case

We frame the AI chip opportunity in China across training and inferencing. On training, we frame the market opportunity based on: (1) the annual number of large / mid / small / light models trained per CSP (cloud service provider) and enterprise, and (2) the number of H800 equivalent GPUs required to train a single model. On inferencing, we frame the market opportunity across: (1) 10 key applications, (2) analysing the user base of each application, (3) token consumption and the revenues generated from the

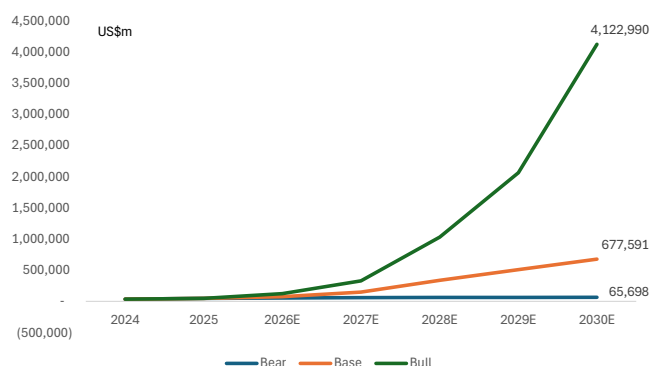
application, (4) GM of each application, and (5) the contribution of AI chips to the COGS. Our Bear / Base / Bull cases for China by 2030E are differentiated by three key variables: (1) number of models trained annually per CSP or enterprise, (2) size of the models being trained annually, and (3) daily token consumption across different AI applications, which is determined by the # of users and daily requests per user.

Exhibit 20: China AI chips TAM: Bear / Base / Bull case in 2030E

2030E	Bear	Base	Bull
China AI Chips TAM			
Opportunity (US\$m)	65,698	677,591	4,122,990
Training - China CSPs	6,250	69,375	418,750
Training - China enterprises	10,523	134,698	841,859
Inferencing - Chatbot	17	419	2,979
Inferencing - Search	86	1,332	8,937
Inferencing - Standard agent task	857	20,934	148,947
Inferencing - Complex agent task	42,842	428,191	2,606,565
Inferencing - Chart generation	15	152	931
Inferencing - Video generation	321	9,040	65,164
Inferencing - AI assistant	23	42	75
Inferencing - AI HR	13	35	75
Inferencing - AI data analysis	50	210	500
Inferencing - AI coding	4,701	13,164	28,208
Daily token consumption (T)	489	4,734	28,616
AI chips (H800-equivalent), k units	2,628	27,104	164,920

Source: Goldman Sachs Global Investment Research

Exhibit 21: China AI chip opportunities in our Bear / Base / Bull case



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 22: China AI chip consumption implied total IT power across our Bear / Base / Bull case by 2030E

	Bear	Base	Bull
Total IT power (kW)	3,127	32,253	196,254
Servers / Supernodes # (k units)	328	3,388	20,615
GPUs # per unit	8	8	8
CPUs # per unit	2	2	2
Computing power per unit (PFLOPS, FP16)	12	12	12
Computing power per GPU (TFLOPS, FP16)	1,500	1,500	1,500
Computing power per GPU (TFLOPS, FP8)	3,000	3,000	3,000
IT power per unit (kW)	10	10	10
TDP per GPU (W)	700	700	700
Coefficient	1.7	1.7	1.7

Source: Goldman Sachs Global Investment Research

Key findings:

(1) Profitability matters: Our TAM assumption requires AI service providers to enjoy meaningful profitability, in which we assume GM at 30% and AI chip costs at 30% of COGS. A meaningful margin would drive the AI application penetration rate and lead to sustainable AI chip investments that would capture the demand.

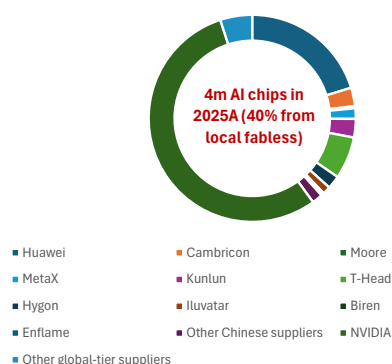
(2) Base case of US\$678bn by 2030E is reasonable based on our cross checks: The base case of US\$678bn by 2030E implies a total IT power requirement of 32GW, i.e. a doubling from the 32GW by end of 2025 in five years ([link](#)). The base case implies daily token consumption at 4,734T (vs. 350T in 2026E), which is consistent with GS' total Chinese AI model daily token consumption forecast of 4,600T by 2030E ([report link](#)).

(3) Local AI chips to enjoy +102% CAGR in 2025-30E in the base case: The base case

of US\$678bn by 2030E implies the need for 27m H800-equivalent AI chips. Assuming local AI chips is 50% of H800 performance on average, we would need 54m local AI chips to achieve the base case market (vs. 1.6m local AI chip shipments in 2025), or +102% CAGR in 2025-30E. The strong market growth would also bring healthy competition through 2030E.

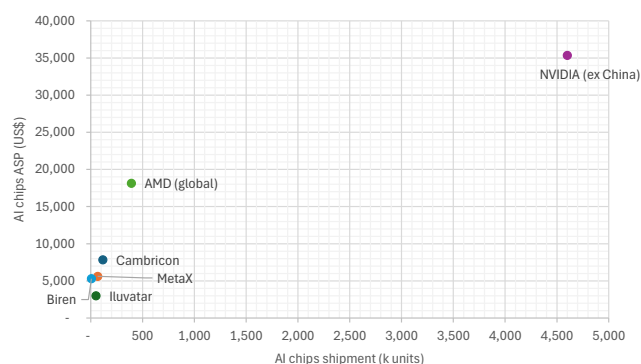
(4) Local SPE, Foundry, and Memory could benefit the most: The AI infrastructure upcycle is driving significant growth in the local AI supply chain, and we expect the following local SPE (e.g. Naura, AMEC), foundry (e.g. SMIC), and memory (e.g. CXMT) players would benefit the most considering they are early participants / contribute significantly to AI chip production, have high technology entry barriers and face fewer local competitors.

Exhibit 23: China AI chip shipments by fabless company in 2025A



Source: IDC

Exhibit 24: AI chips fabless ASP and shipment comparison (2025)



Source: Company data, Goldman Sachs Global Investment Research

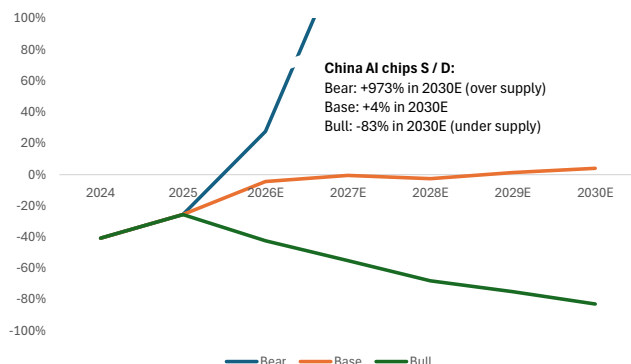
Exhibit 25: China AI chips TAM (base case)

	2024	2025	2026E	2027E	2028E	2029E	2030E	2025-30E CAGR
China AI chips TAM (US\$m)								
Base	36,032	49,231	74,535	149,070	338,796	508,193	677,591	69%
Cambricon	162	908	3,336	12,761	24,118	35,454	44,672	118%
MetaX	156	372	599	1,680	3,880	6,518	9,581	91%
Iluvatar	125	148	394	1,329	2,439	4,230	6,903	116%
Biren	3	34	407	1,375	3,163	4,797	6,601	186%
Other local AI chips	7,005	11,931	44,713	130,222	303,662	455,815	608,592	120%
Global-tier AI chips	28,581	35,837	25,086	1,704	1,534	1,380	1,242	-49%
Unit TAM (k units)								
Base	2,730	4,016	5,884	10,763	21,987	31,086	39,231	58%
Cambricon	26	116	405	1,033	1,859	2,602	3,123	93%
MetaX	24	66	107	274	603	965	1,351	83%
Iluvatar	38	50	107	268	483	821	1,314	92%
Biren	1	7	65	133	303	478	702	155%
Other local AI chips	736	1,368	3,513	8,929	18,626	26,117	32,651	89%
Global-tier AI chips	1,906	2,409	1,686	126	114	102	92	-48%
ASP (US\$)								
Base	13,198	12,259	12,668	13,851	15,409	16,348	17,272	7%
Cambricon	6,252	7,832	8,239	12,358	12,976	13,625	14,306	13%
MetaX	6,508	5,616	5,571	6,128	6,434	6,756	7,094	5%
Iluvatar	3,296	2,986	3,668	4,952	5,051	5,152	5,255	12%
Biren	5,412	5,291	6,314	10,362	10,451	10,031	9,409	12%
Other local AI chips	9,522	8,719	12,729	14,584	16,303	17,453	18,639	16%
Global-tier AI chips	14,995	14,875	14,875	13,500	13,500	13,500	13,500	-2%

Source: Goldman Sachs Global Investment Research, IDC, Company data

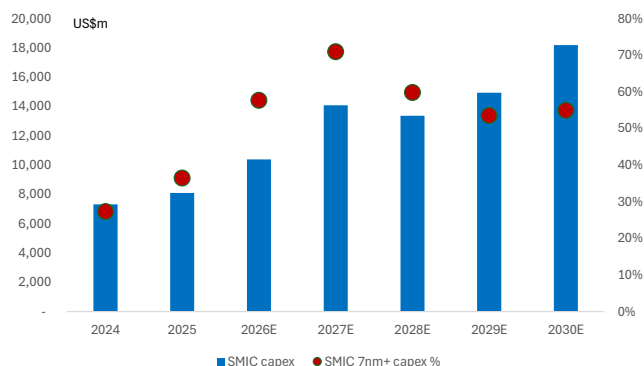
(2) Supply: Capex & yield rate ramp up to capture growth

Exhibit 26: China AI chip supply vs. demand in our Bear / Base / Bull case



Source: Goldman Sachs Global Investment Research

Exhibit 27: SMIC: capex and 7nm+ capex to total capex



Source: Company data, Goldman Sachs Global Investment Research

Key findings:

(1) Supply constraints: Based on our scenario analysis, local foundries need to have an aggressive capacity expansion plan along with continuous yield rate ramp up to catch with the AI chip demand in China. Our bull case shows that more than one local foundry undergoing strong capacity and yield rate ramp up would be required in order to capture the full AI chip opportunity in China.

(2) SPE and material suppliers to benefit: Our scenario analysis shows local foundries would need to maintain capex at a high level through 2030E for advanced node capacity expansion, which would benefit local SPE and material suppliers (e.g. CMP slurry, ceramic heaters, etc). The strong visibility reaffirms our positive view on Naura, AMEC, ACMR, Anji, Kematek.

(3) China market features: We note the supply chain is enhancing overall computing power through various approaches, such as adoption of chiplet architectures and advanced packaging, that allow manufacturers to restrict the use of the most advanced nodes to the primary computing die. By keeping individual die sizes smaller than the integrated SoC, fabrication complexities are reduced. Supernodes which connect multiple AI chips to function as a single unit also significantly boost aggregate system computing power. Software optimizations can also supplement these hardware advancements to further enhance both training and inferencing performance.

Exhibit 28: S/D in our Bear / Base / Bull case

	2024	2025	2026E	2027E	2028E	2029E	2030E	2025-30E CAGR
Demand								
China AI chips (k units)								
Bear	2,730	4,016	4,408	4,269	4,050	3,898	3,804	-1%
Base	2,730	4,016	5,884	10,763	21,987	31,086	39,231	58%
Bull	2,730	4,016	9,764	23,814	66,892	126,099	238,715	126%
Dual die shipment contribution								
Bear	50%	50%	50%	50%	50%	50%	50%	
Base	50%	50%	50%	50%	50%	50%	50%	
Bull	50%	50%	50%	50%	50%	50%	50%	
China AI chips dies (k units)								
Bear	4,095	6,024	6,612	6,403	6,076	5,847	5,706	-1%
Base	4,095	6,024	8,825	16,144	32,980	46,628	58,847	58%
Bull	4,095	6,024	14,645	35,721	100,338	189,149	358,072	126%
Supply								
SMIC 7nm+ capacity (k wpm)	20	35	65	115	155	195	245	48%
Net add (k wpm)		15	30	50	40	40	50	
For AI chips	50%	50%	50%	50%	50%	50%	50%	
Capacity for AI chips (k wpm)	10	18	33	58	78	98	123	48%
Annual capacity for AI chips (k wafers)	121	210	390	690	930	1,170	1,470	48%
# of die per wafer (units)	100	97	94	91	89	86	83	-3%
Yield rate	20%	22%	23%	26%	39%	47%	50%	
# of die post yield rate (k units)	2,427	4,481	8,440	16,058	32,110	47,222	61,223	69%
7nm capacity addition (k wpm)	10	15	30	50	40	40	50	
Investment per 10k wpm capacity (US\$m)	2,000	2,000	2,000	2,000	2,000	2,000	2,000	
SMIC revenues (US\$m)	8,030	9,327	11,694	14,392	18,422	22,435	27,321	24%
YoY	27%	16%	25%	23%	28%	22%	22%	
SMIC capex (US\$m)	7,326	8,102	10,396	14,084	13,368	14,943	18,198	18%
SMIC capex ratio	91%	87%	89%	98%	73%	67%	67%	
SMIC 7nm+ as a % of total capex	27%	36%	58%	71%	60%	54%	55%	
S/D								
Bear	-41%	-26%	28%	151%	429%	708%	973%	
Base	-41%	-26%	-4%	-1%	-3%	1%	4%	
Bull	-41%	-26%	-42%	-55%	-68%	-75%	-83%	

Source: Company data, Goldman Sachs Global Investment Research, IDC

Advanced packaging ecosystem

Exhibit 29: Advanced packaging ecosystem



Source: Company data, Goldman Sachs Global Investment Research

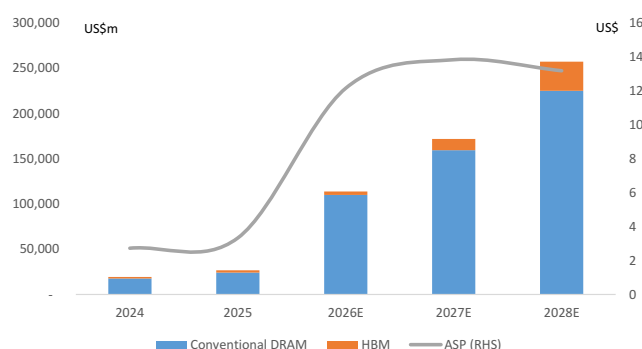
Part 4: China DRAM TAM

(1) Demand: US\$257bn by 2028E; AI to drive Server DRAM growth

Framework: We frame China DRAM demand via eight key applications across consumer electronics (PC, smartphones), infrastructure (server, networking), and automotive. Based on end device shipments of each application in China and DRAM content per device, we estimate China's total DRAM consumption. We benchmark to global DRAM ASP and calculate China DRAM value TAM.

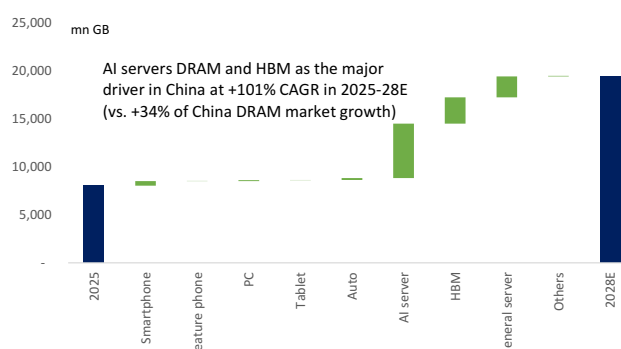
Key findings: We expect China DRAM demand to grow at 50% CAGR in 2026-28E to US\$257bn by 2028E, driven by strong AI server shipments, leading HBM to grow at +188% CAGR in 2026-28E to US\$32bn by 2028E. We expect growing AI foundation models and applications to drive the AI penetration rate, offering sustainable growth for the AI infrastructure supply chain.

Exhibit 30: China DRAM value TAM and ASP trend



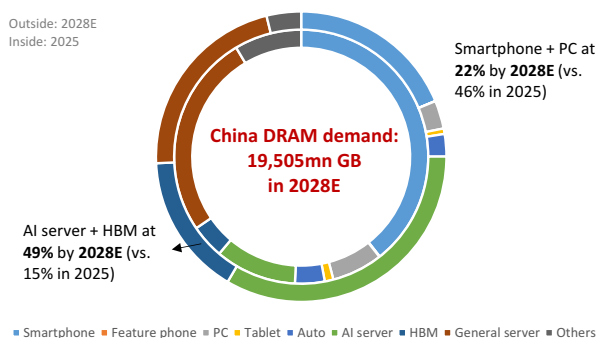
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 31: China DRAM market waterfall chart



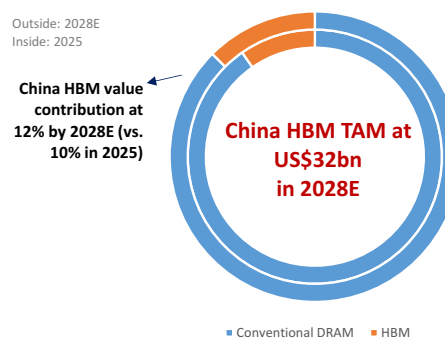
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 32: China DRAM volume TAM by applications: 2025 vs 2028E



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 33: China HBM global value market share: 2025 vs. 2028E



Source: Company data, Goldman Sachs Global Investment Research

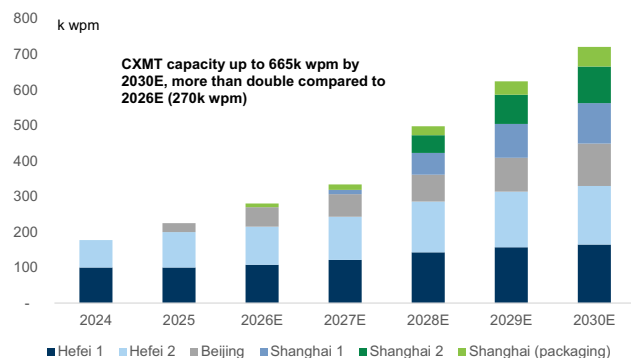
Exhibit 34: China DRAM value TAM

	2024	2025	2026E	2027E	2028E	2026-28E CAGR
China demand by type (US\$m)	19,297	26,822	113,919	171,918	257,315	50%
Conventional DRAM	17,646	24,220	110,046	159,501	225,209	43%
HBM	1,651	2,601	3,873	12,417	32,105	188%
Shipment (mn GB)	7,073	8,050	9,365	12,419	19,505	44%
Conventional DRAM	6,837	7,703	8,818	11,201	16,419	36%
HBM	235	347	547	1,218	3,086	138%
ASP (US\$/GB)	2.7	3.3	12.2	13.8	13.2	4%
Conventional DRAM	2.6	3.1	12.5	14.2	13.7	5%
HBM	7.0	7.5	7.1	10.2	10.4	21%

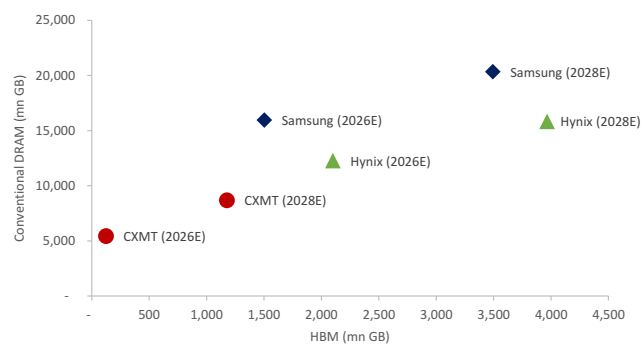
Source: Company data, Goldman Sachs Global Investment Research

(2) Supply: CXMT vs. China market demand

We expect CXMT's capacity to grow from 270k wpm in 2026E to 447k wpm in 2028E, or average annual capex to increase from Rmb60bn (US\$9bn) in 2024-25 to Rmb84bn (US\$12bn) in 2026-30E, supporting its total DRAM supply to grow at +34% CAGR in 2026-30E to 9,837mn GB, and HBM supply to grow at +207% CAGR in 2026-28E to 1,179mn GB. We expect the growing capacity, yield rate ramp up, and product specification upgrades (e.g. HBM toward HBM3 / 3E) will enable domestic suppliers to better capture China's strong market growth and meet rising demand from fabless companies looking to diversify their supplier networks to mitigate macro uncertainties (e.g. pandemics, geopolitical tensions) and secure supply to meet robust AI demand.

Exhibit 35: CXMT capacity expansion trend

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 36: DRAM global shipment: CXMT vs Samsung vs Hynix

Source: Goldman Sachs Global Investment Research

Exhibit 37: China DRAM volume TAM: Supply vs Demand

	2024	2025	2026E	2027E	2028E	2026-28E CAGR
China demand by application (mn GB)	7,073	8,050	9,365	12,419	19,505	44%
Smartphone	2,908	3,171	3,200	3,418	3,655	7%
Feature phone	4	5	3	3	3	-5%
PC	456	521	515	561	611	9%
Tablet	85	88	95	107	119	12%
Auto	233	304	415	447	486	8%
AI server	586	836	1,238	2,596	6,529	130%
HBM	235	347	547	1,218	3,086	138%
General server	1,602	2,090	2,589	3,325	4,266	28%
Other (Networking, CE, Industrial, etc)	963	689	763	744	750	-1%
Supply by vendors (mn GB)						
CXMT	2,191	4,111	5,536	7,382	9,837	33%
Samsung	13,323	14,556	17,462	19,674	23,841	17%
Hynix	9,389	11,638	14,373	16,632	19,791	17%
Micron (including Elpida)	7,195	8,628	10,859	13,134	15,063	18%
Supply by vendors - HBM (mn GB)						
CXMT			125	687	1,179	207%
Samsung	604	564	1,503	2,500	3,494	52%
Hynix	794	1,550	2,102	2,701	3,968	37%
Supply/Demand						
CXMT vs China demand	-69%	-49%	-41%	-41%	-50%	
CXMT HBM vs China HBM demand			-77%	-44%	-62%	

Source: Company data, Goldman Sachs Global Investment Research

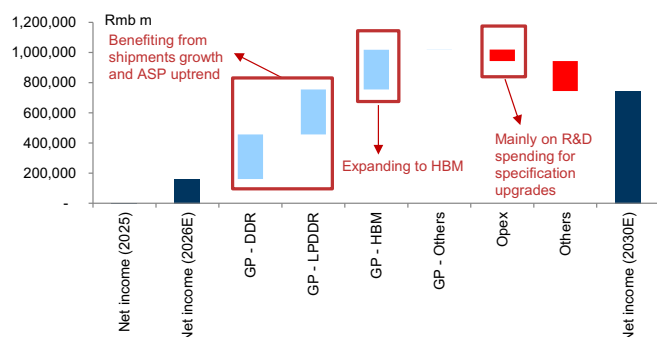
Part 5: CXMT (688825.SS): AI and diversification to drive growth ahead; TP at Rmb129; Initiate at Buy

CXMT is a leading DRAM IDM player in China, covering the design and manufacturing of DRAM and serving major local CSPs and consumer electronics brands. We forecast NI to grow at +47% CAGR in 2026–30E, driven by: (1) strong AI demand in China, driving both servers DRAM and HBM demand, where we expect CXMT's HBM revenues to grow at +166% YoY in 2026–30E, (2) customers' diversifying their supplier networks to better secure supply, especially critical for consumer electronics where standard memory supply is being squeezed by the industry's pivot toward AI centric production. We expect CXMT's conventional DRAM to grow at +34% CAGR in 2026–30E. We expect capacity to more than double in 2030E vs. 2026E towards 665k wpm, with avg. annual capex increasing up to Rmb84bn (US\$12bn) in 2026–30E from Rmb50bn (US\$7bn) in 2022–25A. Initiate at Buy with TP at Rmb129 (implying 24x 2027E P/E vs. +77% 2027–28E avg. EPS).

Key debates: One of the key debates centres on its valuation. We believe that CXMT's scarcity, product mix upgrades, and being a beneficiary of both AI and customers' diversifying suppliers, should lead to its EPS outpacing the industry, thereby supporting its valuation. We note Memory stocks usually trade at a discount to other Semis given they are cyclical; however, CXMT currently trades at 10x 2027E PE against +77% 2027–28E avg. EPS, or 0.13x PEG (vs. Cambricon/Montage/AMEC's at 1.12x/1.46x/2.21x), a significant discount compared to global peers. Our TP implied 24x 2027E P/E, or 0.31x, narrows the discount to other China Semis. Furthermore, CXMT's close collaboration with leading customers, coupled with China's strong commitment to domestic Semis development, enhances CXMT's long-term growth visibility. We forecast China's DRAM market to be at US\$257bn by 2028E, which suggests CXMT's total revenues at ~50% of China's DRAM market (vs. ~20–30% in 2024–25A).

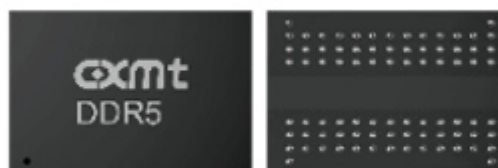
Valuation: Our target price is based on a 2030E discounted P/E. Our target P/E multiple is derived from peers' trading P/E relative to their forward-year EPS YoY. Our TP implies 24x 2027E P/E with NI YoY at 77% in 2027-28E. **Key downside risks:** fiercer-than-expected market competition; weaker-than-expected market demand; unstable geopolitical situation.

Exhibit 38: CXMT's NI waterfall chart 2025-30E
DRAM shipment growth and price uptrend



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 39: CXMT's DDR5



Source: Company data

(1) HBM (High Bandwidth Memory)

HBM (high bandwidth memory) is a specialized, high-performance 3D-stacked SDRAM (synchronous dynamic random-access memory) architecture designed to solve the 'Memory Wall' (the performance bottleneck where processor speeds outpace memory bandwidth) in AI workloads. Key characteristics of HBM include:

- **3D vertical stacking:** HBM stacks up multiple (e.g. 8, 12, or 16) DRAM dies vertically on top of a base logic die.
- **Through-Silicon Vias (TSVs):** The stacked dies are interconnected vertically using thousands of microscopic copper pillars called TSVs. These TSVs pass through the silicon layers, drastically reducing physical distance and latency compared to traditional wire bonding.
- **Ultra-wide interface:** Traditional DDR5 uses a 64-bit bus, and GDDR6 uses a 32-bit bus per channel. In contrast, HBM (up to HBM3E) utilizes a 1024-bit wide interface. The next-generation HBM4 standard doubles this to a 2048-bit interface.
- **System-in-Package (SiP) integration:** HBM are co-located with the host processor on a custom silicon interposer inside the same physical package. This tight coupling minimizes signal degradation and power consumption.

Why AI needs HBM: During AI training and inference, the processor (e.g. GPU, CPU) must constantly load large amounts of model parameters from memory and process them. Without HBM's terabytes-per-second bandwidth, real-time text generation would slow down. HBM has become the indispensable backbone of AI infrastructure because it's an ideal solution for the memory wall. By stacking DRAM dies vertically and integrating them directly with the processor on a shared silicon interposer, HBM delivers

the massive data throughput, high capacity, and power efficiency required to train and run trillion-parameter AI models without wasting the processor's compute cores, enhancing the overall computing efficiency.

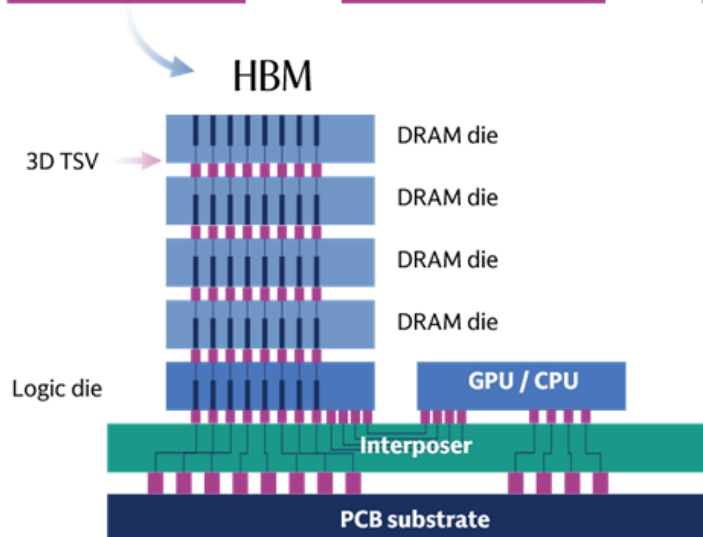
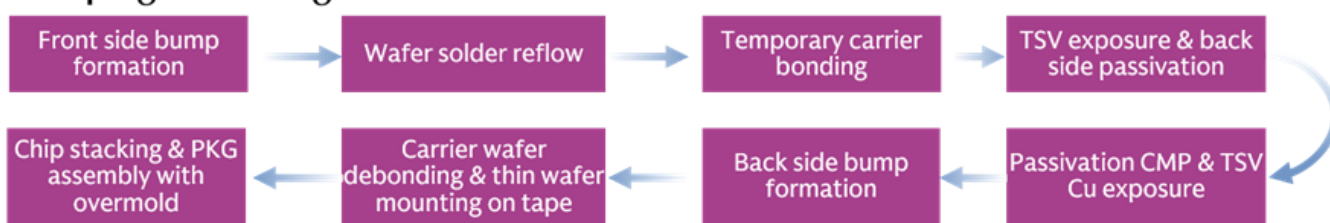
CXMT's HBM in expansion: We expect CXMT to continue to develop and enhance the performance of its HBM products, with contribution projected to begin in 4Q26E. By partnering with the local semiconductor ecosystem, CXMT is working to overcome the key bottlenecks, such as front-end DRAM chip manufacturing, high-precision TSV, logic die in advanced node, thermal management and reliability. We see rising HBM demand in China, driven by AI, and CXMT's HBM should facilitate development of the local AI ecosystem. We expect CXMT's HBM revenue mix to increase from 2% in 2026E to 27% in 2030E.

Exhibit 40: HBM manufacturing process and key bottlenecks

Fab



Bumping & stacking



Key bottlenecks

Front-end DRAM manufacturing

Lack of EUV lithography
low wafer-sort yields limiting the volume of usable dies available for stacking

Through-Silicon Via (TSV) Precision

Any minor defect in TSV alignment or filling ruins the entire multi-die stack, destroying the economic viability of the batch

Advanced node logic die

Lack of advanced node capacity in China caps the logic die shipments and performance

Thermal management and reliability

Excessive heat prevents the chips from maintaining consistent operating speeds and compromises long-term durability

Source: Company data, Data compiled by Goldman Sachs Global Investment Research

(2) Earnings estimates

Revenues: We forecast +86% CAGR in 2025-30E, driven by: **(1) strong AI demand in China**, driving both servers DRAM and HBM demand, where we expect CXMT's HBM revenues to grow at +166% YoY in 2026-30E, **(2) customers' diversifying their suppliers to better secure supply**, especially for consumer electronics under strong AI

demand. We expect CXMT's conventional DRAM to grow at +34% CAGR in 2026-30E. We expect capacity to more than double in 2030E vs. 2026E toward 665k wpm, to support and enable CXMT to capture the strong demand. We expect product mix upgrade to support blended ASP and revenue growth, and model HBM revenue contribution to go up to 27% by 2030E (vs. 2% in 2026E). **GM:** Up from 41% in 2025 to 82% in 2030E, mainly on product mix upgrade along with ASP staying high on strong AI demand. **Opex ratio:** Improving from 27.4% in 2025 to 7.9% in 2030E on growing revenue scale and higher operational efficiency, with R&D expense at a +56% CAGR to support the product development of DDR, LPDDR, and HBM. **NI:** +231% CAGR in 2025-30E, driven by rising revenue scale and an improvement in profitability. We expect **CCC days** to improve from 189 days in 2025 to 84 days in 2030E, on better inventory and accounts payable days, driven by the DRAM shipment ramp-up. We forecast **ROE** to increase to 36% in 2030E from 4% in 2025, mainly on improving asset turnover and net margins driven by product mix upgrade towards LPDDR6, DDR5, and HBM and DRAM price staying high. We believe **FCF** should improve in the coming years, driven by higher operating cash flow as earnings scale up.

Exhibit 41: CXMT P&L summary

Rmb m	1Q26	2Q26E	3Q26E	4Q26E	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Income statement													
Revenues	50,800	65,500	89,494	114,961	8,287	9,087	24,178	61,799	320,756	661,563	878,714	1,103,832	1,369,487
GP	40,212	53,127	74,345	96,386	(259)	(175)	1,349	25,334	264,070	556,549	731,962	913,178	1,122,724
OP	37,175	45,136	65,216	84,660	(3,820)	(7,248)	(6,024)	8,422	232,188	497,008	652,878	815,489	1,014,534
Net income	24,762	30,280	46,184	60,147	(8,328)	(16,340)	(7,145)	1,875	161,373	361,256	472,573	595,020	744,511
EPS (Rmb)	0.41	0.50	0.69	0.90	(0.16)	(0.30)	(0.13)	0.03	2.50	5.40	7.07	8.90	11.13
Margins													
GM	79.2%	81.1%	83.1%	83.8%	-3.1%	-1.9%	5.6%	41.0%	82.3%	84.1%	83.3%	82.7%	82.0%
OPM	73.2%	68.9%	72.9%	73.6%	-46.1%	-79.8%	-24.9%	13.6%	72.4%	75.1%	74.3%	73.9%	74.1%
NM	48.7%	46.2%	51.6%	52.3%	-100.5%	-179.8%	-29.6%	3.0%	50.3%	54.6%	53.8%	53.9%	54.4%
Ratios													
Opex ratio	6.0%	12.2%	10.2%	10.2%	43.0%	77.8%	30.5%	27.4%	9.9%	9.0%	9.0%	8.9%	7.9%
Tax rate	6.8%	8.0%	8.0%	8.0%	0.0%	0.0%	0.0%	7.2%	7.8%	9.0%	9.5%	10.0%	10.5%
YoY													
Revenues	719%	na	438%	na	na	10%	166%	156%	419%	106%	33%	26%	24%
GP	4835%	na	na	na	na	na	na	1778%	942%	111%	32%	25%	23%
OP	na	na	na	na	na	na	na	na	2657%	114%	31%	25%	24%
Net income	na	na	na	na	na	na	na	na	8507%	124%	31%	26%	25%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 42: CXMT's cash flow statement

Rmb m	2024	2025	2026E	2027E	2028E	2029E	2030E
Cash flow statement							
Net income	(7,145)	1,875	161,373	361,255	472,573	595,020	744,511
Minority interest add-back	(1,906)	5,269	48,045	89,077	117,328	138,566	162,070
Depreciation and amortization add-back	15,803	26,356	31,209	33,392	62,851	77,963	83,414
(Increase)/decrease in working capital	(8,262)	(7,668)	(24,713)	(22,702)	13,079	(33,800)	14,136
Other operating cash flow items	8,407	10,687	(37)	-	-	-	-
Cash flow from operating	6,897	36,520	215,876	461,022	665,832	777,749	1,004,131
Capital expenditure	(71,230)	(49,739)	(45,066)	(81,275)	(123,342)	(99,797)	(72,471)
Other investment cash flow items	(1,186)	(37,084)	(1,500)	(1,500)	(1,500)	(1,500)	(1,500)
Cash flow from investing	(72,415)	(86,823)	(46,566)	(82,775)	(124,842)	(101,297)	(73,971)
Dividends and interests paid	(3,204)	(3,077)	-	-	-	-	-
Change in common stock	4,138	2,422	57,919	-	-	-	-
Increase/(decrease) in short-term debt	22,728	(5,053)	(3,000)	(3,000)	(3,000)	(3,000)	(3,000)
Increase/(decrease) in long-term debt	6,171	22,832	(1,000)	(1,000)	(1,000)	(1,000)	(1,000)
Other financing cash flow items	47,381	28,997	-	-	-	-	-
Cash flow from financing	77,214	46,121	53,919	(4,000)	(4,000)	(4,000)	(4,000)
Net change in cash	11,325	9,292	223,230	374,248	536,989	672,452	926,160
FCF	(64,332)	(13,219)	170,811	379,748	542,489	677,952	931,660
Ratio							
Capex to revenue	295%	80%	14%	12%	14%	9%	5%

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 43: CXMT's balance sheet

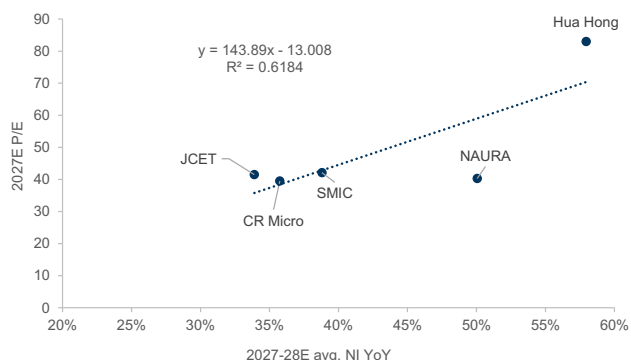
Rmb m	2024	2025	2026E	2027E	2028E	2029E	2030E
Balance Sheet							
Cash and equivalents	42,699	51,990	275,220	649,467	1,186,457	1,858,909	2,785,070
Accounts receivable	1,846	1,524	12,537	20,088	28,060	32,423	35,113
Inventory	18,550	28,568	47,531	73,307	71,435	106,161	110,180
Other current assets	4,920	28,904	28,904	28,904	28,904	28,904	28,904
Current assets	68,014	110,986	364,192	771,767	1,314,856	2,026,398	2,959,266
Net PP&E/Fixed assets	153,132	183,024	198,629	248,138	310,205	333,585	324,169
Net intangibles	5,153	4,278	4,067	3,940	3,864	3,818	3,791
Other long-term assets	45,299	38,497	38,497	38,497	38,497	38,497	38,497
Non-current assets	203,585	225,799	241,193	290,576	352,567	375,901	366,457
Total assets	271,599	336,785	605,385	1,062,343	1,667,423	2,402,298	3,325,724
Accounts payable	4,658	6,687	11,950	22,575	41,754	47,044	67,887
Short-term debt	34,712	29,659	26,659	23,659	20,659	17,659	14,659
Other current liabilities	17,821	16,497	16,497	16,497	16,497	16,497	16,497
Current liabilities	57,191	52,842	55,105	62,731	78,910	81,199	99,043
Long-term debt	95,994	118,825	117,825	116,825	115,825	114,825	113,825
Other long-term liabilities	14,157	11,017	11,017	11,017	11,017	11,017	11,017
Non-current liabilities	110,151	129,843	128,843	127,843	126,843	125,843	124,843
Total liabilities	167,342	182,685	183,948	190,574	205,752	207,042	223,886
Common stock	57,771	60,193	118,112	118,112	118,112	118,112	118,112
Retained earnings	(38,522)	(36,650)	124,723	485,978	958,551	1,553,571	2,298,082
Other common equity	22,171	33,211	33,211	33,211	33,211	33,211	33,211
Total common equity	41,420	56,754	276,046	637,302	1,109,874	1,704,894	2,449,405
Minority interest	62,837	97,346	145,391	234,467	351,796	490,362	652,433
Total equity	104,257	154,100	421,437	871,769	1,461,670	2,195,257	3,101,838
BVPS	0.73	0.98	4.28	9.53	16.59	25.49	36.62
Cash conversion cycle							
Account receivable days	17	10	8	9	10	10	9
Inventory days	234	236	245	210	180	170	160
Net payable days	66	57	60	60	80	85	85
Cash conversion cycle	185	189	193	159	110	95	84
Ratios							
ROE (common equity)	-19%	4%	97%	79%	54%	42%	36%
ROA	-3%	1%	34%	43%	35%	29%	26%
Net cash per share (Rmb)	(1.56)	(1.67)	2.03	7.61	15.70	25.81	39.72
Total liabilities to total assets	62%	54%	30%	18%	12%	9%	7%
Dupont analysis							
Asset turnover	0.1	0.2	0.7	0.8	0.6	0.5	0.5
Leverage (assets to equity)	6.0	6.2	2.8	1.8	1.6	1.4	1.4
Net margin	-30%	3%	50%	55%	54%	54%	54%

Source: Company data, Goldman Sachs Global Investment Research

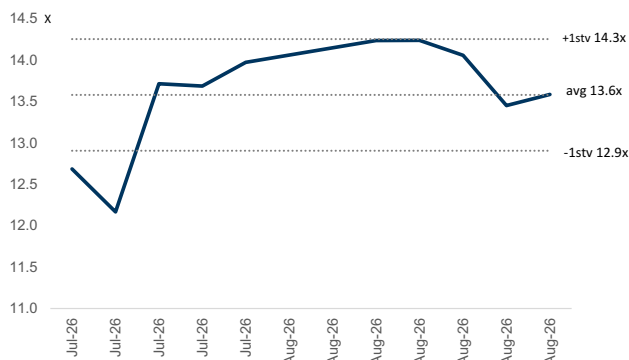
(3) Valuation and Risks

We derive our 12-month target price of Rmb129 based on a discounted P/E methodology to capture the company's long-term growth, which is consistent with our Greater China Tech coverage. Our 2030E target P/E multiple of 16.6x is based on the peers' correlation between forward-year trading P/E and earnings growth, with CXMT's 2030-31E average net income YoY at 21%. We apply the 16.6x target P/E to 2030E EPS and discount it back to 2027E at a COE of 12.7%. **Initiate at Buy.** When CXMT enters a normalized growth phase, our TP implies 12x 2030E PE against 21% avg. EPS YoY in 2030-31E, or 0.6x, within the peers' range of 0.3x-1.2x PEG. We note Memory stocks usually trade at a discount compared to other Semis given they are cyclical. Once maturity is reached and the cycle restarts, we could revisit our valuation methodology and compare its near-term PB vs. ROE to global-tier peers. CXMT currently trades at 88-94% discount compared to local SPE and fabless names, showing attractive valuation. We believe the stock at our valuation is not stretched as our TP implies a 2027E PE at 24x, or 0.31x PEG, and is at a 72% to 86% discount to

Cambricon/Montage/AMEC.

Exhibit 44: Our target P/E multiple is derived from peers' correlation between P/E and forward-year NI YoY growth

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 45: CXMT's 12M forward P/E

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 46: CXMT and peers' PEG ratio

Company	CXMT	Company	Samsung	SK Hynix	Micron	SMIC	NAURA	Hua Hong	JCET
Ticker	688825.SS	Ticker	005930.KS	000660.KS	MU	0981.HK	002371.SZ	1347.HK	600684.SS
Rating	Buy	Rating	Buy*	Buy	Neutral	Buy	Buy	Buy	Neutral
2030E TP implied P/E	12	2027E trading P/E	4	3	6	42	40	83	42
2030-31E avg. NI YoY	21%	2027-28E avg. NI YoY	33%	15%	52%	39%	50%	58%	34%
PEG	0.6	PEG	0.1	0.2	0.1	1.1	0.8	1.4	1.2

* on Asia CL

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 47: CXMT's memory peers comp table

Company	Ticker	Rating	Market cap US\$m	2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E
CXMT	688825.SS	Buy	547,323	22	10	8	8507%	124%	31%	419%	106%	33%	72%	75%	74%
Peers															
Samsung	005930.KS	Buy	1,235,563	6	4	3	579%	47%	17%	116%	29%	14%	53%	61%	62%
SK Hynix	000660.KS	Buy	850,748	5	3	3	417%	40%	14%	268%	42%	16%	76%	79%	77%
Micron	MU	Neutral	944,838	8	6	NA	239%	51%	NA	146%	43%	NA	80%	83%	NA
Montage	688008.SS	Buy	37,114	61	47	34	84%	30%	36%	51%	36%	41%	46%	48%	47%
Gigadevice	603986.SS	Buy	37,426	50	42	32	222%	18%	31%	80%	20%	21%	33%	32%	35%
Longsys	301308.SZ	Not Covered	25,502	10	25	39	1299%	-43%	-38%	157%	18%	-3%	3%	8%	19%
Biwin	688525.SS	Not Covered	16,925	18	22	14	3076%	-40%	97%	214%	-15%	67%	5%	6%	NA
Average			449,731	23	21	21	845%	15%	26%	147%	25%	26%	42%	45%	48%

Data as of Aug 14, 2026. Estimates for not covered (NC) names are based on consensus.

Source: Company data, Goldman Sachs Global Investment Research, Refinitiv Eikon

Exhibit 48: CXMT and peers' current share price implied PEG ratios

Company	CXMT	Cambricon	Montage	AMEC
Ticker	688825.SS	688256.SS	688008.SS	688012.SS
Rating	Buy	Buy	Buy	Buy
2027E PE	9.9	68.7	48.5	58.0
2027-28E EPS YoY	77%	61%	33%	26%
Ratio	0.13	1.12	1.46	2.21

Source: Company data, Goldman Sachs Global Investment Research

M&A framework: We assign an M&A rank of 3 (indicating a low probability of the company being acquired), given it is currently in a rapid growth stage driven by its DRAM business, and therefore, we do not include any M&A component in our price target. **Key downside risks: (1) Fiercer-than-expected market competition:** We see global peers expanding their DRAM capacity and developing next-generation products with stronger performance, and fiercer-than-expected market competition could weigh on CXMT's revenue and earnings growth. **(2) Weaker-than-expected market demand:** The rising end demand for DRAM leads to price hikes, which is one of our main

assumptions that drives CXMT's margin expansion in the coming years. If DRAM demand is weaker than expected, it would bring downside risk to CXMT's DRAM shipments and margin. **(3) Unstable geopolitical situation:** CXMT is expanding to global brand customers' supply chains, which would drive the company's shipments growth. The unstable geopolitical situation may lead to restrictions for global customers to procure CXMT's memory products, bringing downside risk to CXMT's shipments and revenue growth.

Exhibit 49: CXMT's discounted P/E

Rmb m	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	2031E
Milestones	DDR & LPDDR			Migration to HBM			Capacity expansion and rising DRAM end demand			
Revenue	8,287	9,087	24,178	61,799	320,756	661,563	878,714	1,103,832	1,369,487	1,663,927
YoY		10%	166%	156%	419%	106%	33%	26%	24%	22%
Gross profit	-259	-175	1,349	25,334	264,070	556,549	731,962	913,178	1,122,724	1,347,781
Gross margin	-3.1%	-1.9%	5.6%	41.0%	82.3%	84.1%	83.3%	82.7%	82.0%	81.0%
OPEX	3,561	7,073	7,372	16,912	31,882	59,541	79,084	97,689	108,189	129,786
YoY		99%	4%	129%	89%	87%	33%	24%	11%	20%
Opex ratio	43.0%	77.8%	30.5%	27.4%	9.9%	9.0%	9.0%	8.9%	7.9%	7.8%
Operating profit	-3,820	-7,248	-6,024	8,422	232,188	497,008	652,878	815,489	1,014,534	1,217,995
YoY		na	na	na	2657%	114%	31%	25%	24%	20%
Operating margin	-46%	-80%	-25%	14%	72%	75%	74%	74%	74%	73%
Non-op	-5,350	-11,975	-3,025	-721	-5,007	-2,137	-1,053	-393	-1,594	-1,594
Pre-tax profit	-9,170	-19,223	-9,049	7,701	227,180	494,871	651,825	815,096	1,012,940	1,216,400
Tax	1	2	2	557	17,762	44,538	61,923	81,510	106,359	158,132
Tax rate	0%	0%	0%	7%	8%	9%	10%	10%	11%	13%
Minority	(843)	(2,885)	(1,906)	5,269	48,045	89,077	117,328	138,566	162,070	194,624
Net profit	(8,328)	(16,340)	(7,145)	1,875	161,373	361,256	472,573	595,020	744,511	863,644
EPS (Rmb, diluted)	(0.2)	(0.3)	(0.1)	0.0	2.5	5.4	7.1	8.9	11.1	12.91
YoY		na	na	na	8507%	124%	31%	26%	25%	16%
TP implied P/E						24	18	14	12	10

2030E target P/E	16.6
Target multiple x EPS	185
Discounted back to 2027; TP (Rmb)	129.0
Implied 2027 P/E	24

COE assumption	
Beta	1.5
Risk free	3.0%
Market risk premium	6.5%
COE	12.7%

Source: Company data, Goldman Sachs Global Investment Research

CXMT is a leading DRAM IDM player in China, covering the design and manufacturing of DRAM chips and serving major local CSPs and consumer electronics brands. We expect strong net income growth ahead, driven by (1) rising DRAM shipments supported by strong demand from AI infrastructure; (2) DRAM ASP uptrend in the coming years on tight supply globally, supporting the company's revenue growth and margin expansion; (3) DRAM capacity expansion, supported by capex uptrend in the coming years; (4) product line expansion towards HBM; (5) long-term supply agreement signed with major customers, leading to a higher visibility for the company's revenue growth. **We are Buy-rated on CXMT.**

Valuation: Our 12-month target price of Rmb129 is based on a 16.6x 2030E P/E, discounted back to 2027E at a COE of 12.7%. Our target multiple is derived from peers' correlation between forward-year trading P/E and earnings growth.

Key downside risks: Fiercer-than-expected market competition; Weaker-than-expected market demand; Unstable geopolitical situation.

Disclosure Appendix

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We, Allen Chang, Verena Jeng, James Schneider, Ph.D., Shuhei Nakamura, Alexander Duval, Giuni Lee, Daiki Takayama, Yifan Hu, Ting Song, Chao Wang, Anmol Makkar, Kaho Otake, Anant Jakhar, Taeyong Lee and Al Wang, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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MINDCRAFT: OUR THEMATIC DEEP DIVES

Artificial Intelligence



Humanoid Robots



China Self-Sufficiency



Space



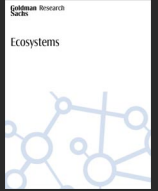
Navigating China Internet



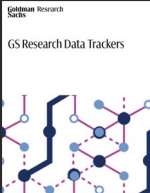
The Ecosystem of EV



Ecosystems



GS Research Data Trackers



Autonomous driving



China Biopharma



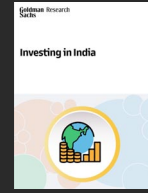
China Consumer



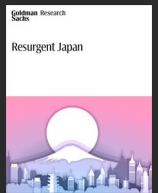
CHIPS Act Impact



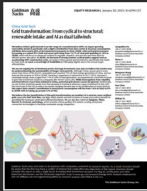
Investing in India



Resurgent Japan



China Grid Tech



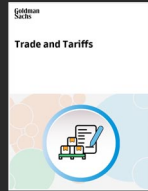
China Property



Supply Chain Shifts



Trade and tariffs



Optical Networking



Top of Mind



Carbonomics



Market Cycles



Tracking the Consumer



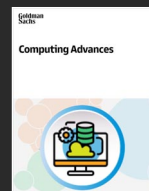
Korea Value in Action



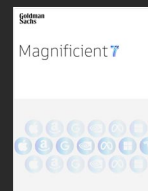
Cybersecurity and Defense



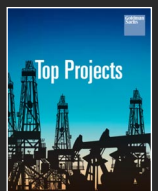
Computing Advances



Magnificent 7



Top Projects



China Going Global



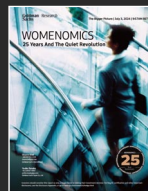
Future of Energy



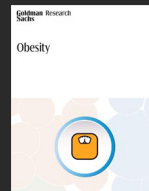
Power Demand



Black Womenomics



Obesity



Geopolitics



Understanding China's Statistics



Clean Hydrogen



Green Capex



ESG

